

## ULGI, Mongolia

WMO: 442140

Lat: 48.9800N

Lon: 89.9383E

Elev: 5627

StdP: 11.94

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -22.9      | -18.0 | -31.5                         | 1.1 | -18.6 | -28.2 | 1.4 | -13.7 | 31.0                  | 15.1 | 26.4 | 15.8 | 1.0                   | 270  | 0.680 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.1                   | 80.7            | 56.1 | 77.9 | 54.8 | 75.3 | 53.7 | 60.0                | 74.3 | 57.7 | 72.2 | 56.0 | 69.8 | 8.9                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 54.5                            | 78.1 | 64.5 | 52.0 | 71.2 | 63.4 | 49.9 | 65.8 | 62.7 | 29.2          | 74.0 | 27.4 | 72.0 | 26.2 | 69.7 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 27.6              | 23.2 | 20.7 | -25.6                      | 85.5 | 5.9                | 3.1 | -29.8                                              | 87.7 | -33.3      | 89.5 | -36.6      | 91.2 | -40.8      | 93.4 |
|                   |      |      | -25.0                      | 61.5 | 5.8                | 4.4 | -29.2                                              | 64.7 | -32.6      | 67.3 | -35.9      | 69.8 | -40.1      | 73.0 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 34.9   | 1.4   | 9.0   | 22.8  | 38.7  | 49.8  | 61.3  | 64.3  | 60.5  | 48.7  | 34.9  | 18.9  | 6.4   |
|                                                                        | DBStd       | 23.39  | 10.05 | 10.65 | 10.49 | 9.23  | 8.03  | 6.02  | 5.05  | 5.57  | 7.62  | 8.71  | 10.52 | 9.87  |
|                                                                        | HDD50       | 6826   | 1507  | 1149  | 844   | 350   | 103   | 5     | 0     | 3     | 111   | 470   | 932   | 1352  |
|                                                                        | HDD65       | 11101  | 1972  | 1569  | 1309  | 790   | 472   | 137   | 73    | 157   | 490   | 933   | 1382  | 1817  |
|                                                                        | CDD50       | 1296   | 0     | 0     | 0     | 9     | 97    | 343   | 445   | 329   | 71    | 2     | 0     | 0     |
|                                                                        | CDD65       | 98     | 0     | 0     | 0     | 0     | 1     | 25    | 53    | 19    | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 855    | 0     | 0     | 0     | 0     | 14    | 212   | 418   | 198   | 13    | 0     | 0     | 0     |
|                                                                        | CDH80       | 115    | 0     | 0     | 0     | 0     | 0     | 23    | 71    | 21    | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 7.4    | 5.9   | 6.1   | 8.2   | 9.5   | 9.3   | 7.8   | 7.1   | 6.6   | 7.1   | 7.3   | 7.4   | 6.4   |
| (15) Precipitation                                                     | PrecAvg     | 4.2    | 0.0   | 0.0   | 0.0   | 0.2   | 0.4   | 1.1   | 1.3   | 0.9   | 0.4   | 0.1   | 0.0   | 0.1   |
|                                                                        | PrecMax     | 6.5    | 0.2   | 0.1   | 0.4   | 0.6   | 1.7   | 6.7   | 5.8   | 3.3   | 1.7   | 0.5   | 0.5   | 0.5   |
|                                                                        | PrecMin     | 2.1    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 1.1    | 0.0   | 0.0   | 0.1   | 0.2   | 0.3   | 1.0   | 0.9   | 0.7   | 0.3   | 0.1   | 0.1   | 0.1   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 30.1  | 36.1  | 53.2  | 67.3  | 75.9  | 83.0  | 85.7  | 82.8  | 75.6  | 61.3  | 48.3  | 32.9  |
|                                                                        |             | MCWB   | 23.9  | 27.1  | 37.9  | 44.7  | 52.1  | 55.2  | 59.0  | 57.6  | 52.5  | 44.3  | 37.0  | 25.5  |
|                                                                        | 2%          | DB     | 24.5  | 31.9  | 48.5  | 61.4  | 71.3  | 79.2  | 81.6  | 78.9  | 70.3  | 56.2  | 43.0  | 28.1  |
|                                                                        |             | MCWB   | 19.0  | 24.3  | 35.2  | 42.2  | 50.1  | 54.1  | 56.7  | 55.4  | 50.0  | 41.7  | 33.3  | 22.0  |
|                                                                        | 5%          | DB     | 20.5  | 28.6  | 44.4  | 57.2  | 67.5  | 76.2  | 78.4  | 75.8  | 66.5  | 52.7  | 38.6  | 24.8  |
|                                                                        |             | MCWB   | 15.9  | 21.9  | 32.3  | 40.1  | 48.0  | 53.0  | 55.8  | 54.3  | 48.4  | 39.8  | 30.4  | 19.6  |
|                                                                        | 10%         | DB     | 16.5  | 25.0  | 40.3  | 53.6  | 64.0  | 73.3  | 75.6  | 72.7  | 62.9  | 49.2  | 34.4  | 21.2  |
|                                                                        |             | MCWB   | 12.6  | 19.1  | 30.0  | 38.4  | 46.0  | 52.0  | 54.6  | 53.1  | 46.6  | 37.7  | 27.2  | 16.7  |
|                                                                        | 0.4%        | WB     | 23.9  | 27.9  | 39.9  | 46.7  | 54.7  | 59.7  | 63.9  | 61.8  | 55.2  | 46.0  | 38.5  | 25.9  |
|                                                                        |             | MCDB   | 29.8  | 35.6  | 50.7  | 63.0  | 72.3  | 76.6  | 78.5  | 77.2  | 70.4  | 56.9  | 48.0  | 32.4  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 19.5  | 24.7  | 35.8  | 43.4  | 51.9  | 56.9  | 60.9  | 59.2  | 52.0  | 42.8  | 34.4  | 22.6  |
|                                                                        |             | MCDB   | 24.5  | 31.6  | 47.0  | 58.5  | 67.8  | 72.5  | 73.4  | 73.7  | 66.1  | 54.8  | 42.9  | 28.2  |
|                                                                        | 5%          | WB     | 16.3  | 22.1  | 32.8  | 41.2  | 49.5  | 55.4  | 58.6  | 56.7  | 49.6  | 40.4  | 30.8  | 20.1  |
|                                                                        |             | MCDB   | 20.7  | 28.6  | 43.9  | 55.2  | 64.0  | 69.5  | 73.1  | 70.6  | 63.8  | 51.3  | 38.5  | 25.1  |
|                                                                        | 10%         | WB     | 13.0  | 19.3  | 30.2  | 39.3  | 47.2  | 54.0  | 56.8  | 54.8  | 47.5  | 38.0  | 27.7  | 17.2  |
|                                                                        |             | MCDB   | 16.9  | 25.2  | 39.9  | 52.0  | 62.2  | 68.5  | 70.3  | 67.8  | 61.4  | 48.8  | 34.7  | 21.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 18.0  | 22.0  | 22.9  | 21.4  | 19.9  | 18.9  | 18.1  | 19.6  | 21.4  | 20.1  | 17.1  | 15.4  |
|                                                                        |             | MCDBR  | 18.8  | 22.7  | 25.3  | 25.1  | 24.4  | 22.8  | 22.4  | 24.3  | 26.4  | 25.2  | 20.1  | 16.8  |
|                                                                        |             | MCWBR  | 15.1  | 17.3  | 16.4  | 13.7  | 12.4  | 8.5   | 8.3   | 9.0   | 12.9  | 15.1  | 13.7  | 13.0  |
|                                                                        | 5% WB       | MCDBR  | 18.8  | 22.2  | 24.1  | 22.7  | 21.6  | 19.7  | 19.4  | 20.5  | 23.8  | 23.5  | 19.7  | 16.7  |
|                                                                        |             | MCWBR  | 15.3  | 17.0  | 16.1  | 13.3  | 12.2  | 8.7   | 9.5   | 10.2  | 12.9  | 14.9  | 14.0  | 13.3  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.233 | 0.260 | 0.306 | 0.352 | 0.346 | 0.350 | 0.347 | 0.328 | 0.292 | 0.277 | 0.254 | 0.235 |
|                                                                        | taud        |        | 2.367 | 2.310 | 2.230 | 2.203 | 2.278 | 2.351 | 2.383 | 2.409 | 2.469 | 2.444 | 2.382 | 2.338 |
|                                                                        | Ebn at Noon |        | 272   | 286   | 287   | 283   | 289   | 288   | 287   | 289   | 291   | 277   | 259   | 253   |
|                                                                        | Edh at Noon |        | 20    | 28    | 36    | 42    | 40    | 38    | 36    | 33    | 28    | 24    | 20    | 18    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 526   | 871   | 1340  | 1737  | 2016  | 2072  | 2007  | 1800  | 1441  | 893   | 532   | 406   |
|                                                                        | RadStd      |        | 21    | 31    | 64    | 91    | 121   | 105   | 115   | 74    | 84    | 70    | 37    | 24    |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |       | N/A     | N/A | N/A    | N/A     | -2.57 | -2.93 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (8 neighbors) |       | N/A     | N/A | -3.60  | N/A     | -1.31 | -1.91 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page